

Embedded Software Intern

Position Summary

We are seeking a highly motivated, passionate, and tech-savvy Embedded Software Intern to join our R&D Engineering team through our campus hiring program. In this role, you will gain hands-on experience working on real-world embedded systems, bridging the gap between hardware and software. You will work closely with senior mentors to develop, test, and debug low-level firmware that powers our next-generation products.

Key Responsibilities

- **Firmware Development:** Assist in designing, writing, and documenting clean, efficient code in C/C++ for microcontrollers and microprocessors.
- **Peripheral Interfacing:** Learn and implement low-level drivers to interface microcontrollers with sensors, actuators, and communication modules using UART, SPI, I2C, or CAN protocols.
- **Testing & Debugging:** Utilize Oscilloscopes, Digital Multimeters, Logic Analyzers, and JTAG/SWD Debuggers to troubleshoot hardware and software integration issues.
- **Hardware Bring-up:** Participate in initial board bring-up activities and validate hardware functionality against schematic designs.
- **Documentation:** Maintain code documentation, testing procedures, and technical requirement specifications.
- **Collaboration:** Participate in code reviews, scrum meetings, and cross-functional sessions with Hardware and QA teams.

Required Skills & Qualifications

- **Education:** Bachelor's or Master's degree in ECE, EE, CSE, or related technical disciplines.
- **Programming:** Strong knowledge of Embedded C/C++. Knowledge of Python for scripting/testing is an advantage.
- **Microcontrollers:** Academic or project experience with ARM Cortex-M, STM32, ESP32, Arduino, or TI MSP430.
- **Core Concepts:** Understanding of Operating Systems, ISRs, Timers, and Memory Management.
- **Hardware Familiarity:** Ability to read basic hardware schematics and component datasheets.
- **Soft Skills:** Strong analytical thinking, eagerness to learn, and effective communication skills.

Preferred / Bonus Qualifications

- Familiarity with FreeRTOS or RTOS concepts.
- Prior experience with Git or other version control systems.

- Hands-on experience through academic projects, robotics clubs, or hackathons
- Familiarity with FreeRTOS or basic Real-Time Operating System concepts.
- Good understanding of RTOS concepts (ThreadX/FreeRTOS)